

EVSWAPP

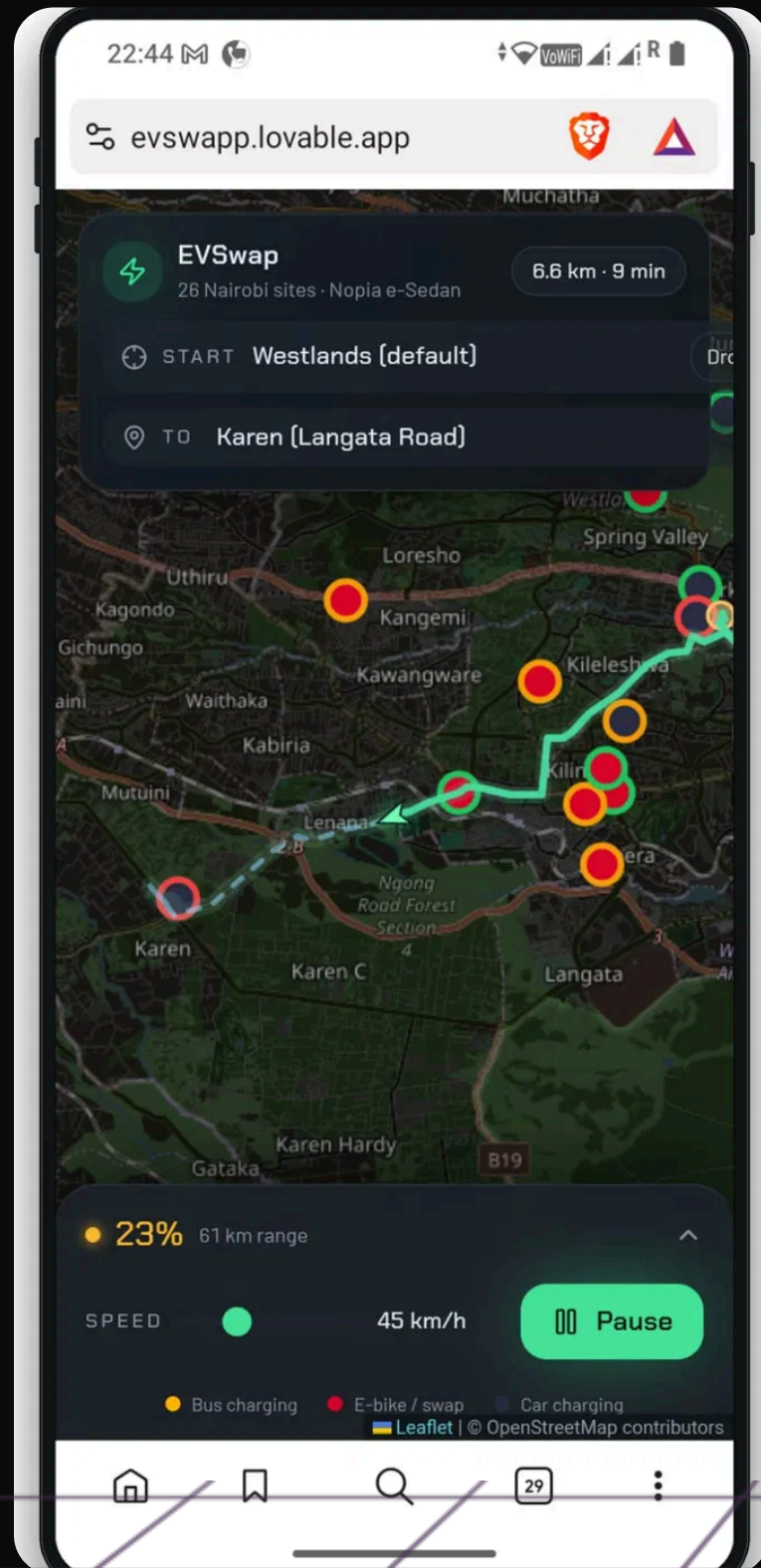
Real-time charge intelligence for
EV drivers in Nairobi



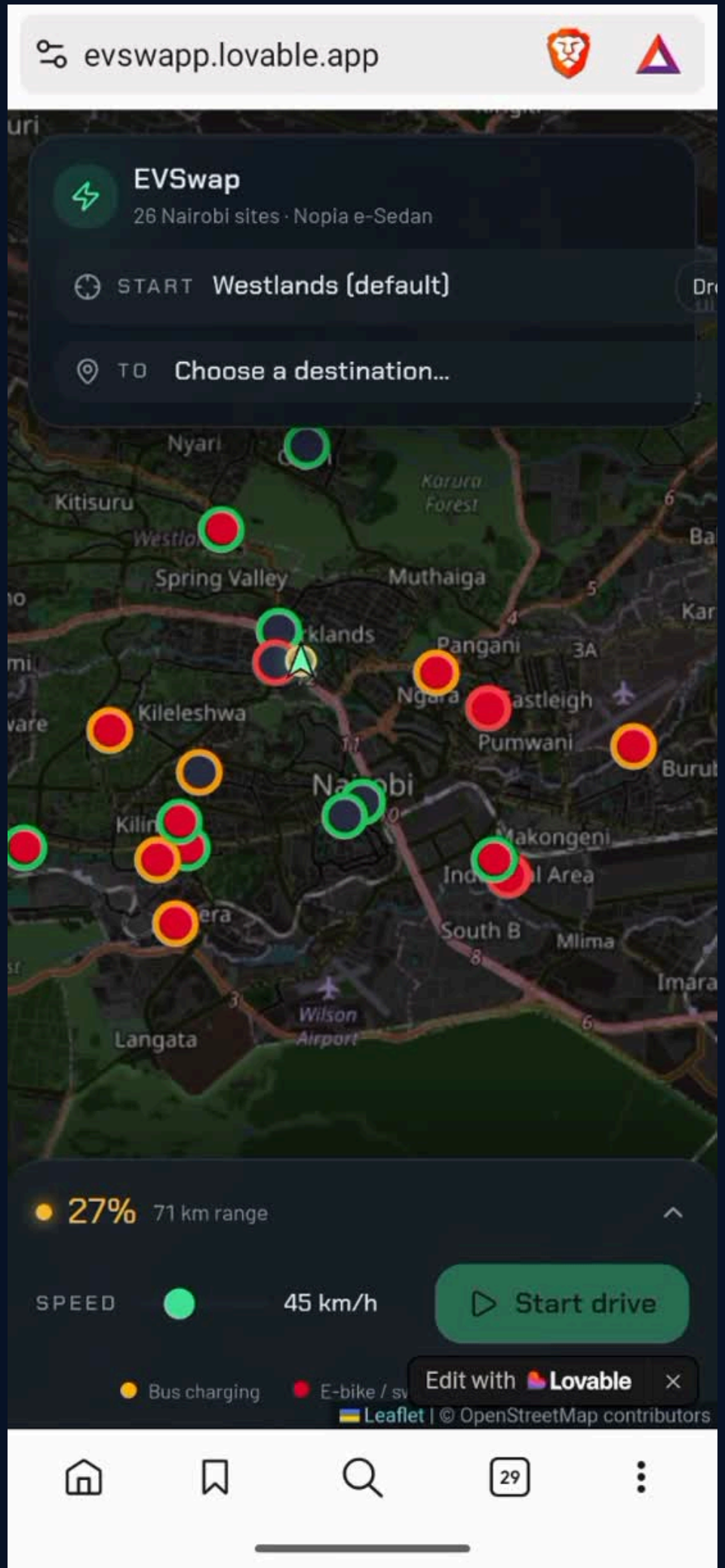
EVSwapp Product Overview

⚡ Simulates a driver's live battery, route, and position and recommends the single best charging or swap point

⚡ Less range anxiety accelerates EV adoption, and better-informed charging behaviour eases pressure on an already uneven grid.



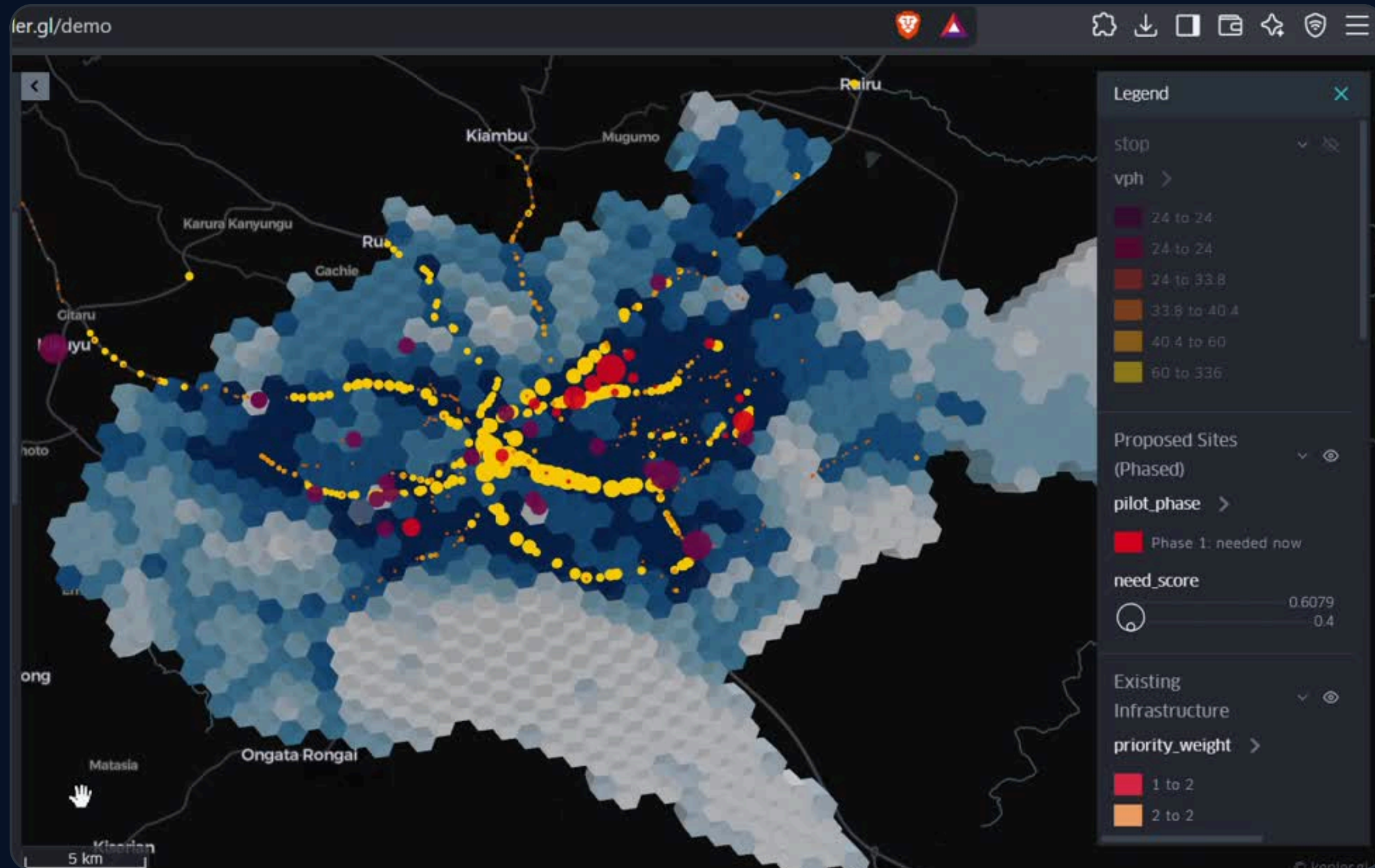
Core Demo - Live Interaction



- 1 Start Drive
- 2 SoC draining live
- 3 AI Copilot feature
- 4 Recommendation appears

What to show: the calm state,
then the moment charge drops
below the safe threshold and the
recommendation card slides in.

Geospatial Demand Layer — Kepler.gl



What this shows

This visualization represents the demand-hotspot intelligence behind the business model — where charging demand is highest relative to existing coverage across Nairobi.

Not part of the live app — this is the site-selection layer sold to real estate developers and entrepreneurs as a separate B2B product.

How The Triage Engine Works

```
const remainingKm = routeDistance - traveled;
const projectedSoC = soc -
  (remainingKm * consumptionRate / capacity) * 100;

if (projectedSoC < SAFE_THRESHOLD) {
  const best = stations
    .filter(s => distanceAhead(s) <= 800)
    .sort(byScore) // reliability, wait time, detour
    [0];

  triggerRecommendation(best);
}
```

- Pure rule-based logic — no external calls
- Recalculates every ~250m of simulated travel
- Safety threshold: 20% projected charge on arrival
- Ranks candidates by reliability, wait time, detour

AI Advisor — LLM-Phrased Recommendations

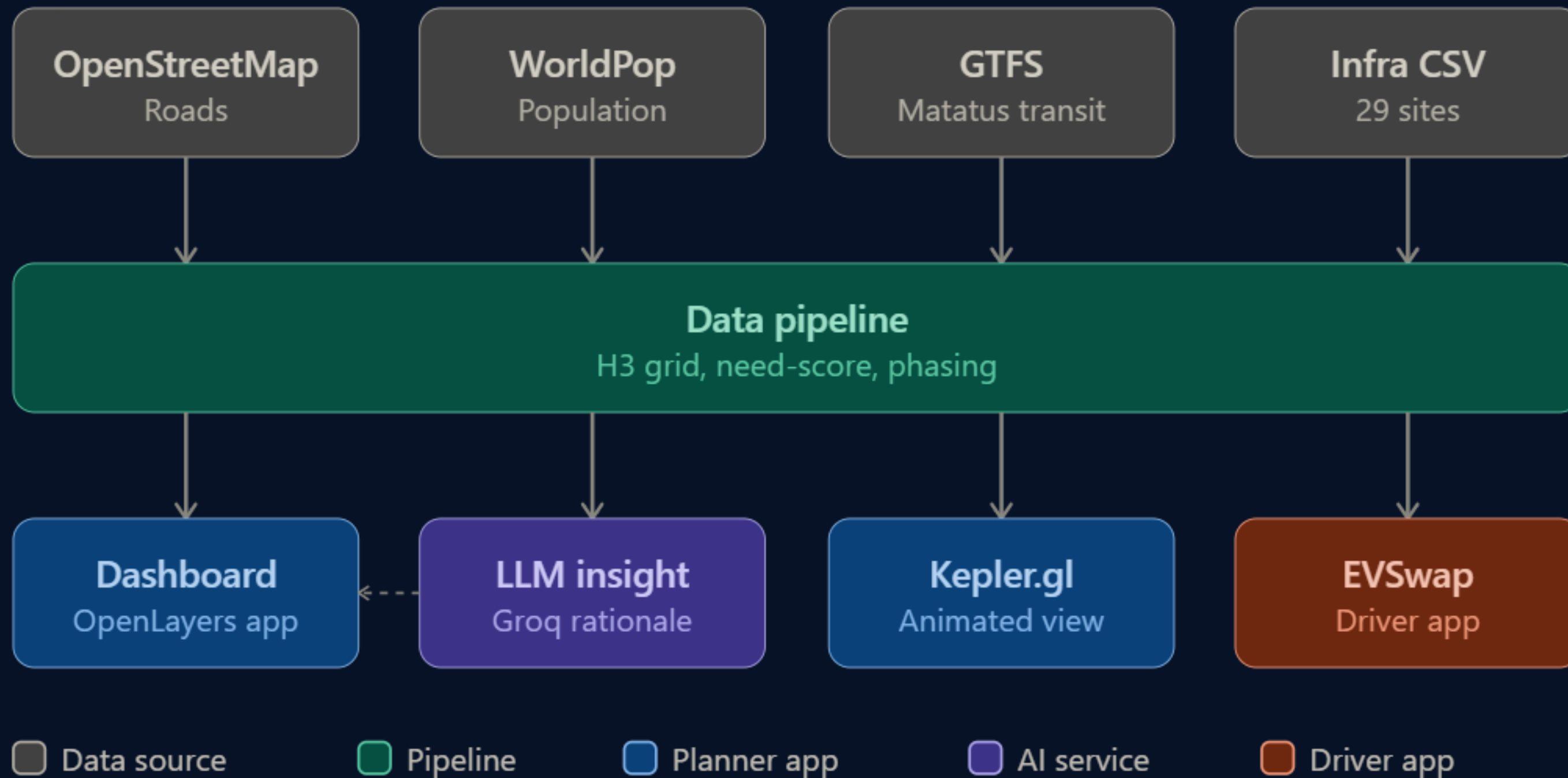
```
const advice = await callLLM({  
  station: best,  
  soc: projectedSoC,  
  location: currentPosition,  
});  
  
// LLM only writes the sentence —  
// station selection stays rule-based
```

Called only when the triage engine actually fires to keep usage minimal.

AI-ADVISED

“Charge drops to ~14% before Karen. Swap in 650m: Kilimani Fast-Charge — 96% reliability, no wait.”

Architecture At A Glance



Try It Yourself

LIVE DEMO

evswapp.lovable.app

CODE REPOSITORY

github.com/PriscahC/EVSwapp

